Language and Thought

Linguistic Relativity, a.k.a. The "Sapir-Whorf Hypothesis"

"No two languages are ever sufficiently similar to be considered as representing the same social reality. The worlds in which different societies live are distinct worlds, not merely the same world with different labels attached."

Sapir (1929)

"We dissect nature along lines laid down by our native language. The categories and types that we isolate from the world of phenomena we do not find there because they stare every observer in the face; on the contrary, the world is presented in a kaleidoscope flux of impressions which has to be organized by our minds—and this means largely by the linguistic systems of our minds. We cut nature up, organize it into concepts, and ascribe significances as we do, largely because we are parties to an agreement to organize it in this way—an agreement that holds throughout our speech community and is codified in the patterns of our language [...] all observers are not led by the same physical evidence to the same picture of the universe, unless their linguistic backgrounds are similar, or can in some way be calibrated."

Whorf (1956)

(The Wikipedia page on this is actually pretty useful, BTW)

Linguistic Relativity

Keep in mind:

Even Sapir and Whorf did not necessarily hold the "strong" relativist position implied in those pull-quotes.

In the first half of the 20th Century, linguistics was really part of anthropology, so treating language as a part of culture seemed more natural than it does to contemporary linguists.

We're going to be talking mainly about *space* and the relationship between *space and time.*

Thinking about space…

Where's the ball?



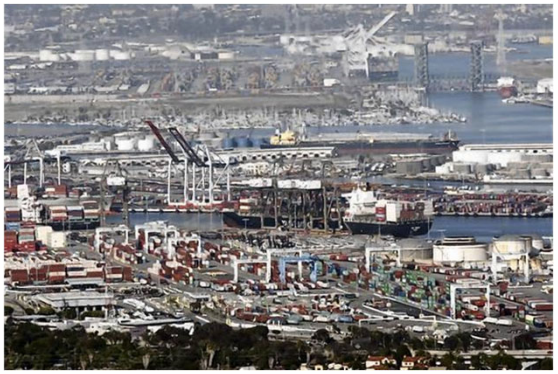
Thinking about space…

(1) Intrinsic: *The ball is at the chair's front.*
(2) Absolute: *The ball is north of the chair.*
(3) Relative: *The ball is to the left of the chair (from my viewpoint).*



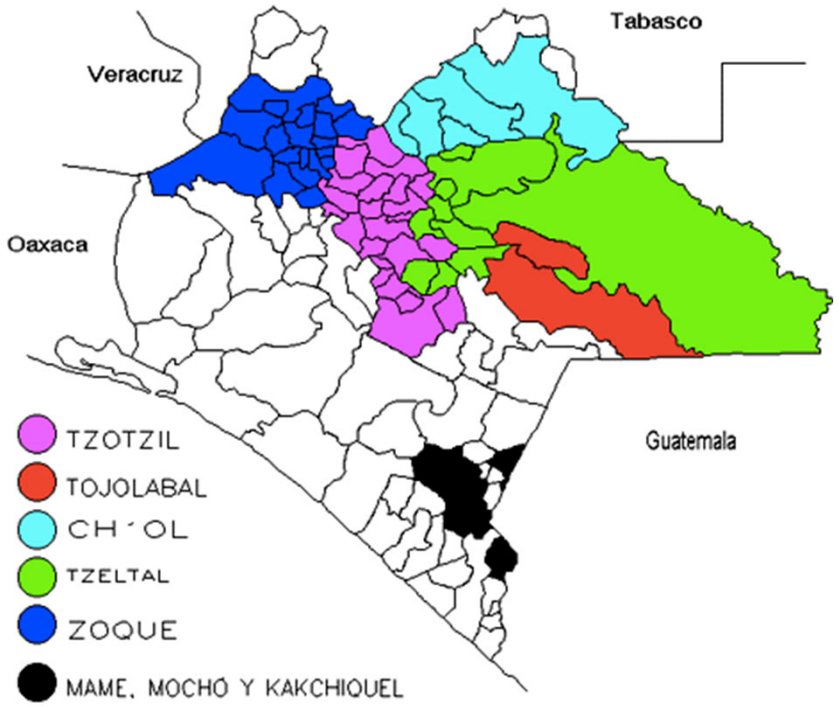






(Levinson et al., 2002, Cognition)

Thinking about space…



It seems natural in most languages to talk about space egocentrically when talking about things that are nearby.

What is to your left or to your right, in front of you or behind you, etc.

Not so in Tzeltal, a Western Mayan language spoken in Chiapas.

In this language (and a small number of others, mostly unrelated to one another, and broadly distributed, e.g., in Bali and Queensland) absolute directions are used.

Raise your east hand if this seems intuitive to you…



Some native Tzeltal speakers defending their land from the vampire squid of neoliberalism.



Beautiful downtown Tenejapa.

Tzeltal has a historically landmark-based system.

Things are either “uphill” or “downhill” from one another, not to the left or right of the speaker or interlocutor.

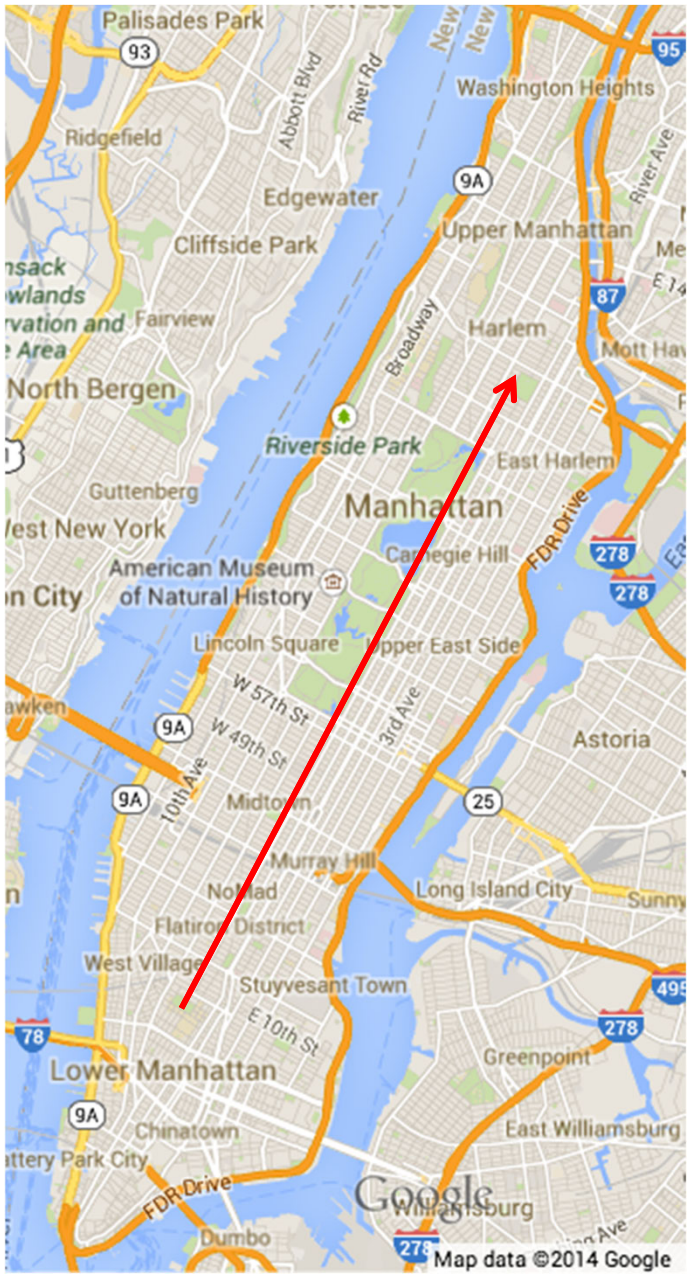
In fact, “uphill” and “downhill” are used as cardinal directions, making this an *absolute* system.

At night, in an alien city, facing a device never seen before (namely a sink with two taps), one Tenejapan asked another, “Which is the hot tap, the uphill (southern) or the downhill (northern) one?”. They maintain a constant sense of absolute orientation, presumably by running a continuous background computation of egocentric heading with respect to abstract bearings, integrating multiple internal and external cues to achieve this.[17] This is the phenomenon that we are trying to capture.

(Levinson et al., 2002, Cognition)

An aside





New Yorkers use “uptown” and “downtown” as if they were cardinal directions all the time, but these don’t work outside of Manhattan.

So that is literally a landmark-based system.

Also, the subway map really has us messed up on “north” and “south,” because Manhattan is on an angle w/r/t magnetic north.

A non-linguistic test for relative vs. absolute coding of space

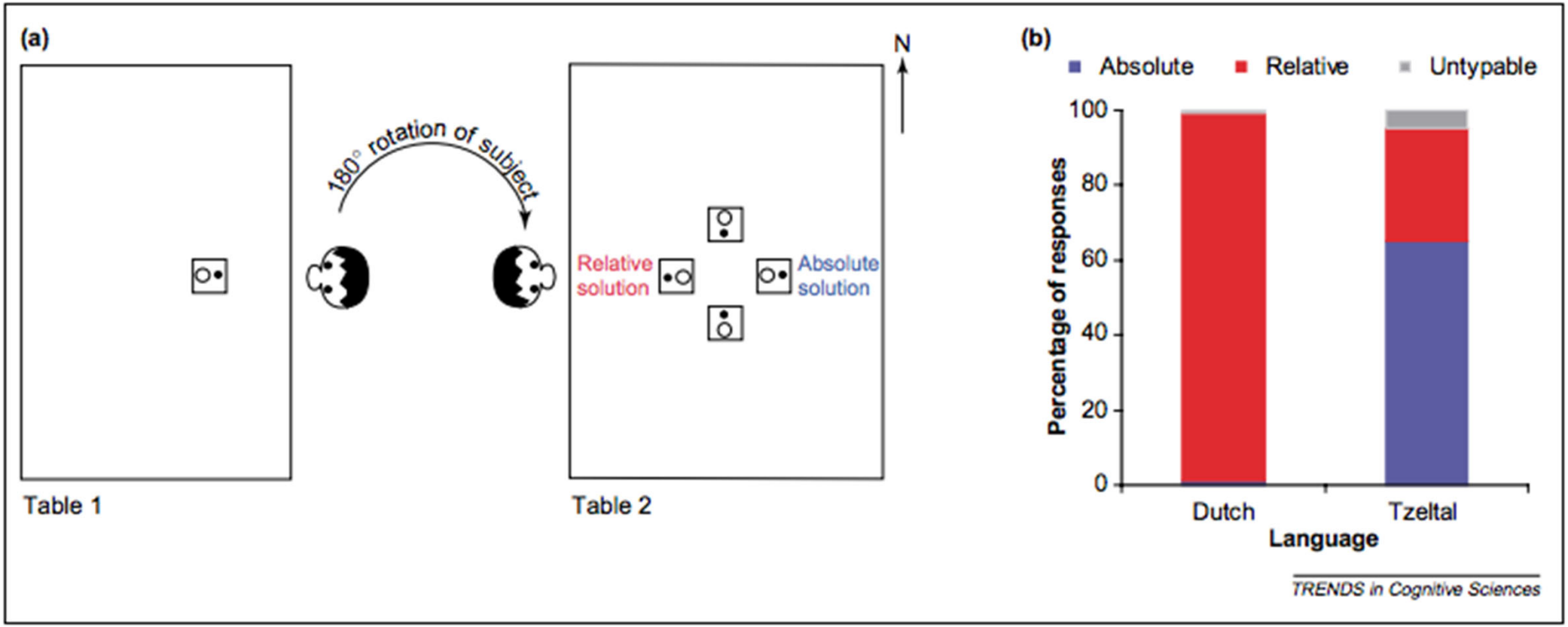


Figure 2. Memory for spatial configuration: 'the chips task'. **(a)** Participants saw on Table 1 a card printed with a large and a small dot arranged with the small dot towards them, away from them, to the left, or to the right. After a 30-s delay, they were rotated through 180° and led to Table 2, where they were asked (in the local language) to identify from a set of four cards the similar or counterpart card to the one they had seen before. Each participant had eight trials, which varied in the arrangement of the dots. Responses that preserved sameness in egocentric coordinates were coded Relative, those that preserved Absolute coordinates were coded Absolute, and other responses were coded 'Untypable'. **(b)** The results for just two languages, Dutch and Tzeltal. The Relative/Absolute trend matches the preferred linguistic FoR: Dutch responses were overwhelmingly Relative whereas Tzeltal responses were overwhelmingly Absolute. The proportionately larger Tzeltal inconsistency can be attributed to the fact that the data are from an unschooled peasant population. However, if we look for 'consistent coders', that is, coders who give the same response on 6 out of 8 trials, then over 80% of Tzeltal speakers are 'Absolute thinkers' (Adapted from [13], pp. 159–160, by permission of Cambridge University Press).

Majid et al. (2004, TiCS)

You can get really fancy with this.

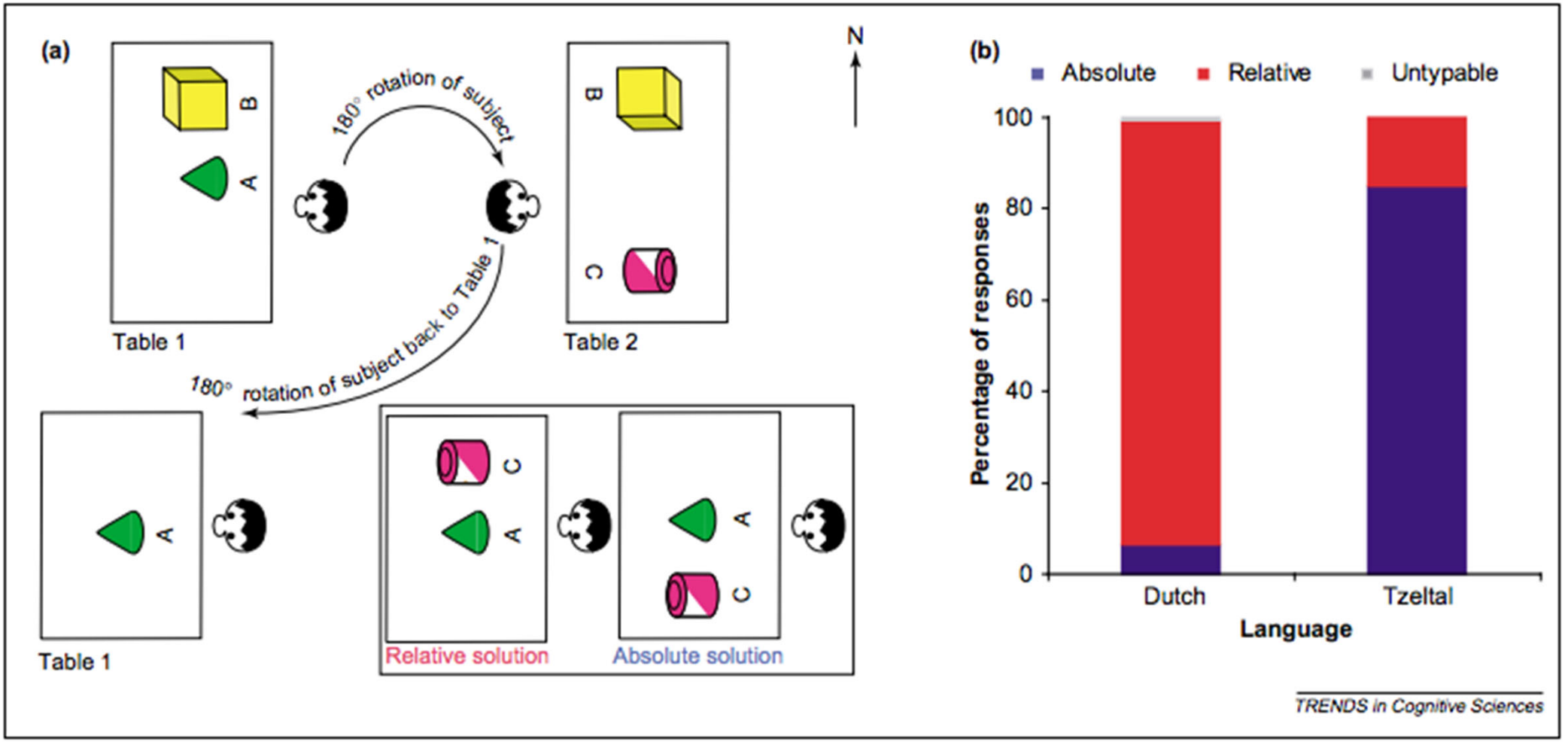


Figure 4. Spatial reasoning: making a transitive inference. **(a)** Transitive inferences (e.g. if A is bigger than B, and B is bigger than C, then A is bigger than C) can be drawn on the basis of non-linguistic arrays. In this experiment, participants saw a cube and a cone arranged in a particular configuration on Table 1, and were then rotated through 180° to Table 2 where they saw a cube and a cylinder. Finally they were rotated back to Table 1, where the cone was standing alone. Their task was to place the cylinder next to the cone, keeping the location consistent with what they had seen before. There were five trials using the transverse axis. There are two ways to perform this task: one using Relative coding (the cube is to the right of the cone, and the cylinder is to the right of the cube, therefore the cylinder is to the right of the cone), and the other using Absolute coding (the cube is to the south of the cone, and the cylinder is further south of the cube, therefore the cylinder is to the south of the cone). **(b)** Once again, the results for Dutch and Tzeltal show that the Relative/Absolute trend matches the preferred linguistic FoR: Dutch participants gave Relative responses whereas Tzeltal participants gave Absolute responses (Adapted from [13], pp. 162–167, by permission of Cambridge University Press).

Majid et al. (2004, TiCS)

Li and Gleitman:
Inducing landmark-based similarity in English speakers

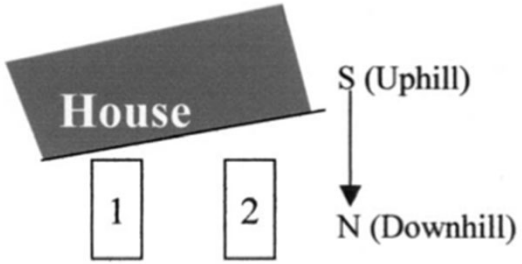


Fig. 4. Test environment for the Tenejapan subjects (following Brown & Levinson, 1993).

Consider the conditions under which the Tzeltal speakers were tested:
Outside, with visible, familiar landmarks.

Typical psychology lab studies are done in windowless rooms.

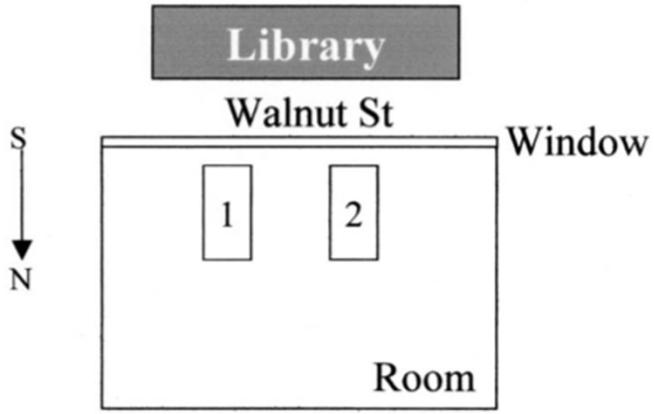


Fig. 5. Test environment for the indoor conditions of Experiment 2a. The University Library is to the south of the tables just as was the house in Fig. 4.

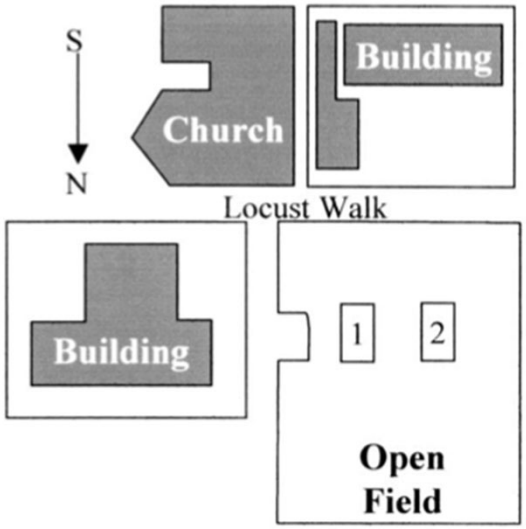


Fig. 6. Test environment for the Outdoor condition of Experiment 2a. Buildings are to the south and southeast of the grassy test area.

Li and Gleitman:

Inducing landmark-based spatial reasoning in English speakers

In the “blinds down” condition, English speakers overwhelmingly use relative position in this task – like the Dutch speakers in prior studies.

With the blinds up and a view of the street, or outdoors, their preference is less pronounced.

In addition to recording subjects' spatial solution to the problem, we also noted whether they asked for clarification during the test or mentioned it during a post-interview. We found that in the Blinds-Down condition (closest to Brown and Levinson's Dutch study), only in 20% of cases did subjects ask for clarification during training ("Should I put them still going the same way, or still going to my left?"). In the other, landmark-rich, conditions, the ambiguity of the task leapt out at the subjects not only in their response styles (the bimodal result shown in Fig. 7) but also in the 70% instance of spontaneous queries during the training phase. Though explicit querying seems to have been rare in the traditional populations, it did sometimes occur (Wilkins, e-mail communication, June 13, 2000; Levy, e-mail communication, November 27, 2000).

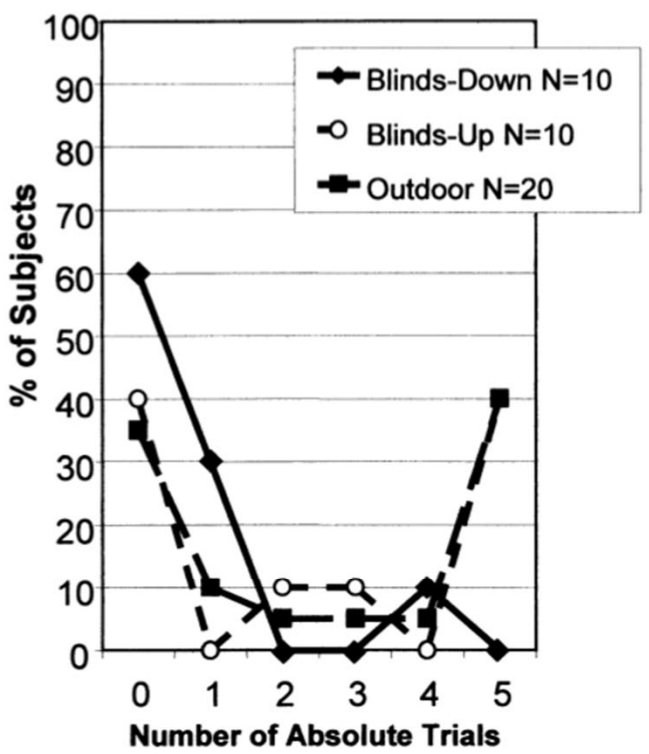


Fig. 7. Proportions of absolute choices in Blinds-Down, Blinds-Up, and Outdoor conditions. Subjects in the Blinds-Down condition predominantly chose the relative response. In the other conditions, subjects were divided in their preference for either all absolute responses or all relative responses.

What does it mean?



What does it mean?

What if the differences between language groups are confounded with differences in testing conditions?

Maybe the bias toward relative spatial reasoning is an artifact of the way psychology tests are typically run in universities…

Of all of our experiments, Li and Gleitman have chosen to replicate the very simplest ('Animals-in-a-row' – see Levinson (in press) for the many converging results from other experiments) and have gone on to simplify it further. The task in essence consists of presenting participants with a row of three animals on a table, rotating the participants 180 degrees, and making them reconstruct the array on another table so that to their satisfaction it matches the first. In our experiment, it was a crucial part of the design that participants' attention was deflected from the *direction* of the stimulus by being required to memorize the *order and identity* of three toy animals drawn from a larger set of four (Levinson, 1996b, p. 114). It was presented as a memory test, first without rotation, then with rotation (and both accuracy of order and direction were coded), and the participant was walked up to 20 m between stimulus and response.

The details of the Li and Gleitman replication vary from the original, including no translation of the participant and no delay after removal of the stimulus and thus a considerably shorter period for retention in memory,[8] and most importantly, the participant's memory task was reduced by presenting the subject with the three animals used in the stimulus, not the full set of four.[9] As we shall show, memory load can make a major difference (see our Experiment 2 below), and in addition when the experiment becomes too transparent participants may second-guess the experimenter's interest in direction rather than order or type of animal. It seems clear from the Li and Gleitman report that many of their participants were second-guessing the experimenter's intentions – they queried the instructions in a way that suggests that direction was clearly at issue. It is always hard to design a task that is matched across schooled and unschooled subject populations, but this should have been a warning that the task was not sufficiently opaque, for what we are interested in is participants' non-reflective utilization of a spatial coding scheme for memory, not what they think the experimenters think they should be doing. Incidentally, for this reason in our original Dutch experiments we used 40 participants of mixed ages (21–77), sex and occupational background, like the participants in our cross-cultural studies.

Levinson and colleagues were not impressed.

They claim that Li and Gleitman made it too transparent that they were testing something to do with rotation by reducing memory load and time/distance between stimulus and test.

Also Li and Gleitman's participants were Ivy League students, who excel at figuring out what the professor wants and then making it happen.

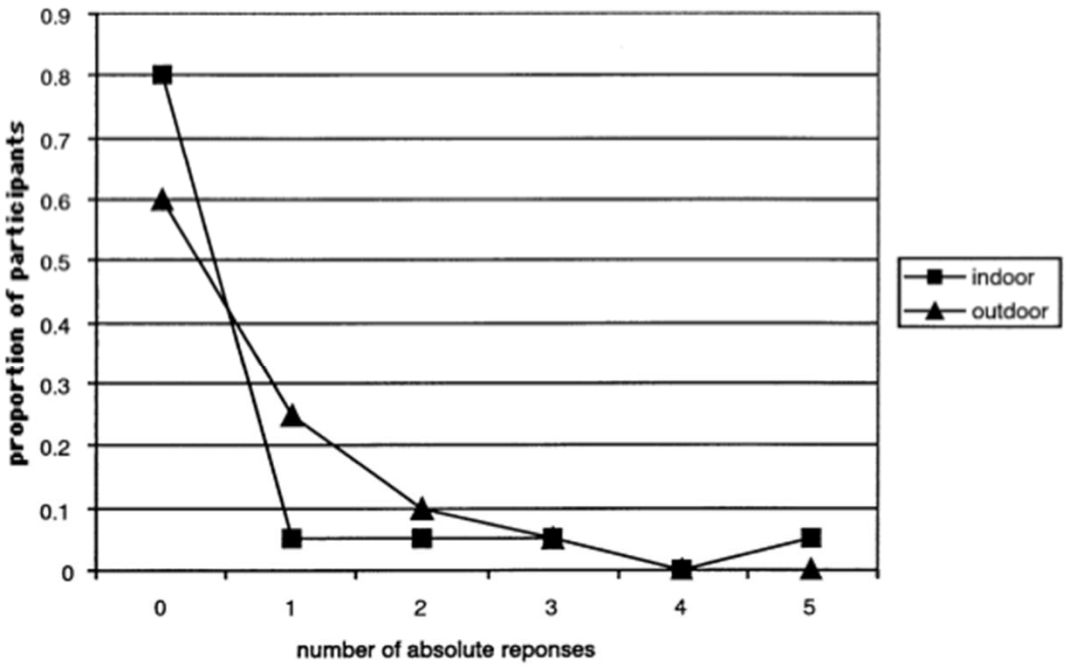


Fig. 4. Direction of animals in the Animals-in-a-row task with Dutch participants: Indoor and Outdoor conditions (a total of five responses from each participant are coded either relative or absolute).

Levinson et al. (2002) re-ran the "animals in a row" experiment with Dutch college students, but including a fourth animal, a longer delay interval (both to increase memory load) and a larger distance between the stimulus and response tables (to reduce the salience of the rotation).

Under these conditions, a bias to use relative framing remains.

What does it mean?

For speakers of languages that use relative spatial frames of reference, the bias to use relative vs. absolute frames is robust to the environment when:

1. There's a significant memory component, and

2. The experimenter's interest in rotation and frames of reference isn't obvious

Can you think of a way in which the importance of "memory" muddies the waters here?

What does it mean?

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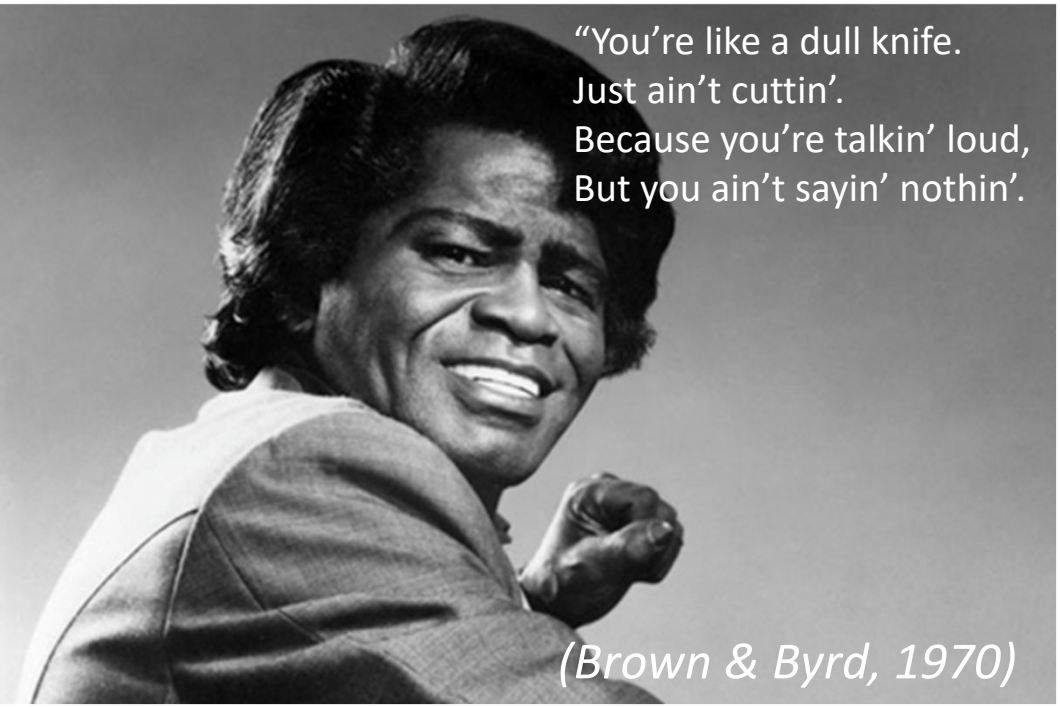
2. The experimenter's interest in rotation and frames of reference isn't obvious

Can you think of a way in which the importance of "memory" muddies the waters here?

What if participants are just *recoding* the list in working memory as a sequence, e.g. "lion, tiger, bear," and then naturally working from left to right?

Why does "left to right" seem like a natural translation of a temporal sequence?

Metaphorical language and “abstract” concepts.



What’s going on here?

How is failing to make a compelling argument like “not cutting?”

Do you literally have to be speaking in a loud voice to be “talking loud” but “saying nothing?”

Obviously, James Brown was a great artist, so we expect him to use evocative language. But how much of language can we safely describe as non-metaphorical?

What is the *point* here?

Is the discussion of Language and Thought a *knotty and convoluted* argument that *generates more heat than light*. In the *following* section, we’ll *pursue* the problem of whether language *has an impact on* how we think about **time** in terms of **space**.

Spatial metaphors for time in different languages

2. Mandarin speakers talk about time vertically more than English speakers do

Both English and Mandarin use horizontal front/back spatial terms to talk about time. In English, we can look *forward* to the good times *ahead*, or think *back* to travails past and be glad they are *behind* us. In Mandarin, front/back spatial metaphors for time are also common. For example, Mandarin speakers use the spatial morphemes *qián* ("front") and *hòu* ("back") to talk about time.

Unlike English speakers, Mandarin speakers also systematically and frequently use vertical metaphors (Chun, 1997a, 1997b; Liu & Zhang, 2009; Scott, 1989; Zhang & Ding, 2003; Zhu, 2006). The spatial morphemes *shàng* ("up") and *xià* ("down") are used to talk about the order of events, weeks, months, semesters, and more. Earlier events are said to be *shàng* or "up", and later events are said to be *xià* or "down". For example, "shàng ge yuè" is last (or previous) month, and "xià ge yuè" is next (or following) month. In one corpus of written Mandarin (Chen, 2007) fully 36% of the spatial metaphors for time were vertical.

(1) SPACE

māo shàng shù
cats climb trees

TIME

shàng ge yuè
last (or previous) month

(2) SPACE

tā xià le shān méi yǒu
has she descended the mountain or not?

TIME

xià ge yuè
next (or following) month

FIG. 2. Example spatial and temporal uses of *up/down* terms *shàng* and *xià* in Mandarin and their English glosses.

(Boroditsky et al., 2010)

A priming experiment

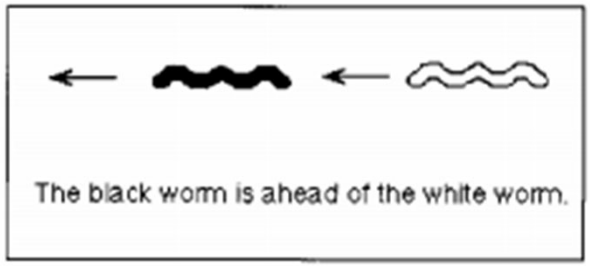


FIG. 3a. Example of a horizontal spatial prime used in Experiments 1 and 3.

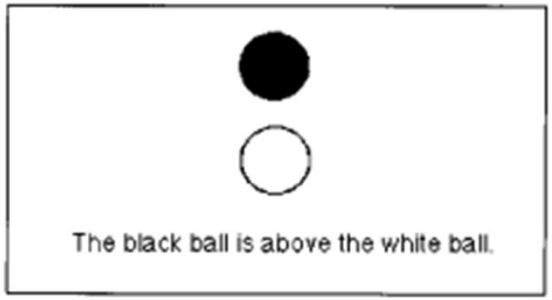


FIG. 3b. Example of a vertical spatial prime used in Experiments 1 and 3.

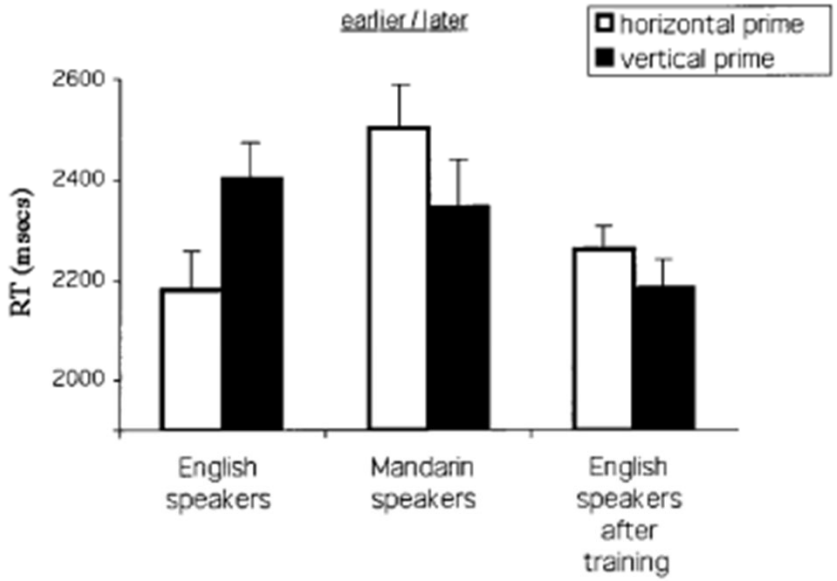


FIG. 4b. Experiments 1 and 3: Response times to purely temporal *earlier/later* questions about time following either a horizontal or a vertical prime are plotted for English speakers, Mandarin speakers, and English speakers who had been trained to talk about time vertically.

Priming with a spatial task (horizontal or vertical) has different effects on temporal processing in the two languages.

Short-term training on vertical metaphors for time (*"Cars were invented above fax machines")* also has an impact.

Boroditsky, 2001)

An alternative explanation: direction of text…

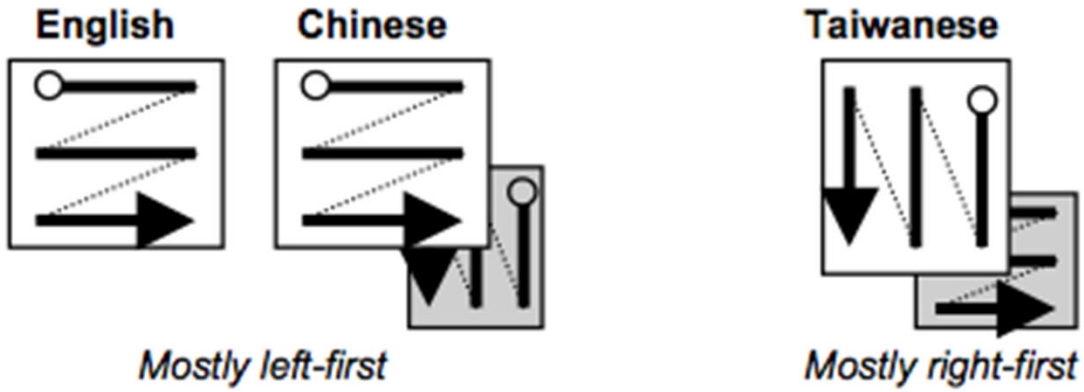


Figure 1: Writing system orientations

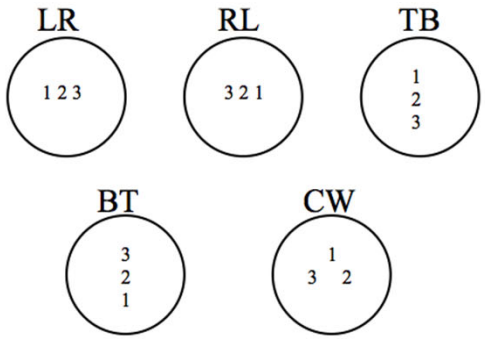


Figure 5: The five observed arrangement patterns

Direction	English	Chinese	Taiwanese	Total
LR	10	26	13	49
RL	0	0	7	7
TB	0	5	13	18
BT	0	1	2	3
CW	0	1	3	4
Total	10	33	38	81

Figure 5: Direction Frequencies by Language

Casasanto & Bottini (2005, Cognitive Science Proceedings)

…so Boroditsky et al. controlled for that…

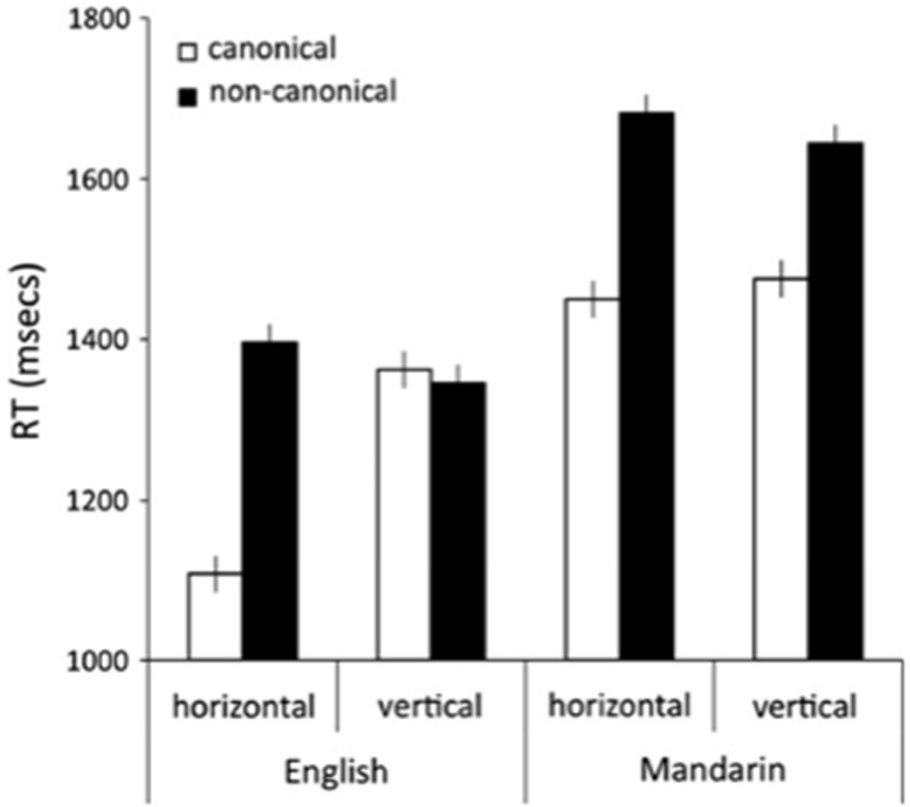


Fig. 1. Native English and native Mandarin speakers' response times. The figure plots by-participants mean response times (in msecs). The error bars represent within-participants standard error (sqrt(MSsc/N)). For the horizontal conditions, the canonical bars show response times in the left-is-earlier key-mapping and the non-canonical bars show response times in the right-is-earlier key-mapping. For the vertical conditions, the canonical bars show response times in the top-is-earlier key-mapping and the non-canonical bars show response times in the bottom-is-earlier key-mapping.

Excluding Chinese readers who reported reading top-to-bottom.

Boroditsky et al. (2010, Cognition)

Task: Push a blue button for events in the past, a red button for events in the future (or the reverse, in different blocks).

Thus, half the time, the “past” is on the left, and the other half it’s on the right.

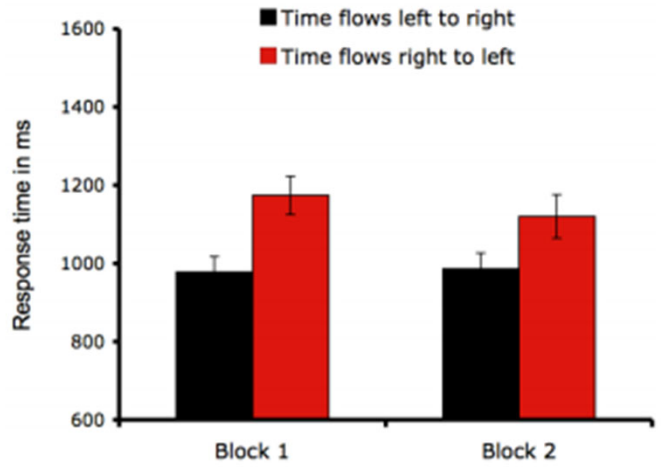


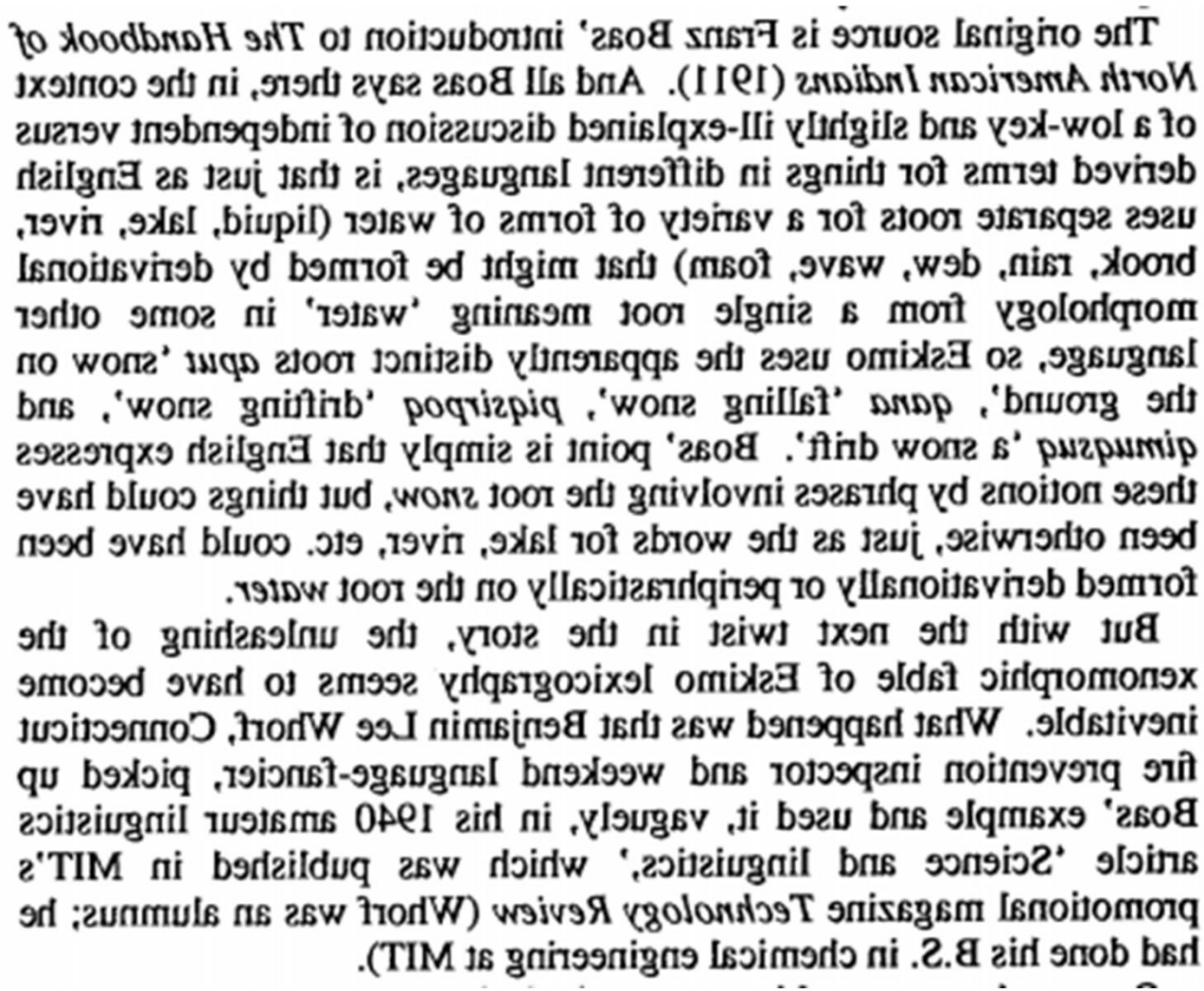
Fig. 1. Results of Experiment 1. When reading standard orthography, participants were faster to judge the temporal reference of phrases when the left-right button press responses were consistent with the rightward flow of time along an implicit mental timeline (left=earlier, right=later). Error bars indicate s.e.m.

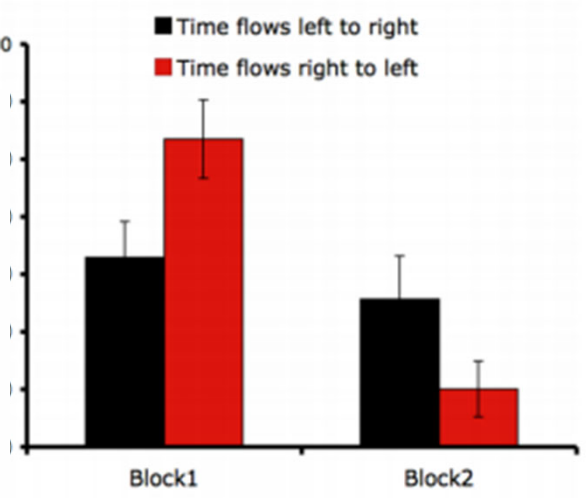
Casasanto & Bottini (2010, Spatial Cognition)

Task: Push a blue button for events in the past, a red button for events in the future (or the reverse, in different blocks).

Thus, half the time, the “past” is on the left, and the other half it’s on the right.

Now ask people to do a few minutes of “mirror reading:”





ment 2. In the first block, the space-time congruity effect was similar iment 1. But by the second block of reading mirror-reversed orthogra- reversed. Participants were faster to judge the temporal reference of it button press responses were consistent with a leftward flow of time imeline (left=later, right=earlier). Error bars indicate s.e.m.

Casasanto & Bottini (2010, Spatial Cognition)

The original source is Franz Boas' introduction to *The Handbook of North American Indians* (1911). And all Boas says there, in the context of a low-key and slightly ill-explained discussion of independent versus derived terms for things in different languages, is that just as English uses separate roots for a variety of forms of water (liquid, lake, river, brook, rain, dew, wave, foam) that might be formed by derivational morphology from a single root meaning 'water' in some other language, so Eskimo uses the apparently distinct roots *aput* 'snow on the ground', *qana* 'falling snow', *piqsirpoq* 'drifting snow', and *qimuqsuq* 'a snow drift'. Boas' point is simply that English expresses these notions by phrases involving the root *snow*, but things could have been otherwise, just as the words for lake, river, etc. could have been formed derivationally or periphrastically on the root *water*.

But with the next twist in the story, the unleashing of the xenomorphic fable of Eskimo lexicography seems to have become inevitable. What happened was that Benjamin Lee Whorf, Connecticut fire prevention inspector and weekend language-fancier, picked up Boas' example and used it, vaguely, in his 1940 amateur linguistics article 'Science and linguistics,' which was published in MIT's promotional magazine *Technology Review* (Whorf was an alumnus; he had done his B.S. in chemical engineering at MIT).

(for those of you who were straining your eyes to read this on the last slide, it's from a classic rant by Pullum (1989, Natural Language and Linguistic Theory) called "The Great *Eskimo Vocabulary Hoax." See if you can guess where he comes down on linguistic relativism…)

*not an OK word for Inuktitut people whose language is the most likely source for the examples, but it's the title of the book, sorry ☹

For next week: Bavelier et al. on cortical plasticity

http://www.nature.com/nrn/journal/v3/n6/abs/nrn848.html